

CSCI 461: Introduction to Big Data - Assignment #2

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a. The MongoDB container running

```
3m 57s base.py 07:09:47 PM
$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
eba863a2cf6d   mongo:7.0                           "docker-entrypoint.s..." 12 seconds ago Up 11 seconds 0.0.0.0:27017->27017/tcp, [::]:27017->27017/tcp
d92bd9b84d1e   portainer/portainer-ce:latest       "/portainer"            3 days ago    Up 21 hours   9000/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->9443/tcp
```

b & c. The database moviesDB created and collection moviesColl after importing

```
~/code/nu/big_data/BigDataAssignment2 main 71 base.py 07:10:53 PM
$ docker exec -it mongo mongosh
Current Mongosh Log ID: 69fa16a854121d348244ba88
Connecting to: mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.8.2
Using MongoDB: 7.0.32
Using Mongosh: 2.8.2

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

To help improve our products, anonymous usage data is collected and sent to MongoDB periodically (https://www.mongodb.com/legal/privacy-policy).
You can opt-out by running the disableTelemetry() command.

-----
The server generated these startup warnings when booting
2026-05-05T16:09:48.144+00:00: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
2026-05-05T16:09:48.144+00:00: Soft rlimits for open file descriptors too low
-----

test> use moviesDB
switched to db moviesDB
moviesDB> show collections
moviesColl
moviesDB>

moviesDB> db.moviesColl.findOne()
{
  _id: ObjectId('69fa168d05e241a8acc65c9b'),
  link: 'https://www.amazon.com/gp/video/detail/amzn1.dv.gti.fcbaefca-e87f-9adb-a197-9f951e3117bb',
  name: 'Ice Airport Alaska',
  description: 'A chronicle of the daily challenges of the airport staff who manage to keep the Ted Stevens Anchorage International Airport (ANC) open, freight flowing, and passengers flying, all despite weather that regularly dips to -38 Fahrenheit and wind speeds that typically exceed 100 mph. Ice Airport Alaska has one or more episodes streaming on Paramount+, streaming with subscription on FuboTV, streaming with subscription on Smithsonian Earth (Via Prime Video), and 3 others. It's a documentary show with 12 episodes over 2 seasons. Ice Airport Alaska is still airing with no announced date for the next episode or season. It has a high IMDb audience rating of 7.8 (48 votes).',
  released_at: '2020-10-26',
  genre: 'Documentary',
  poster: 'https://img.reelgood.com/content/show/0045f9a9-943b-4dda-9a01-384d5bc51f3d/poster-780.jpg',
  streaming_on: 'Paramount+',
  country: 'USA',
  number_of_seasons: '2',
  type: 'TV Show',
  content_rating: 'TV-14',
  imdb_rating: '7.8/10',
  cast_and_crew: [],
  unit_id: '99e5e0f0-3cc0-52e0-a9ed-79dcd7ba0591',
  scraped_at: '02/07/2022 20:36:09'
}
```

d. The command used to import the JSON file

```
24w 55s base.py 07:36:14 PM
$ docker exec mongo mongosh --use moviesDB --collection moviesColl --file /mnt/nu/nu/BigDataAssignment2/moviesColl.json --jsonArray
2026-05-05T16:36:19.725+0000 connected to: mongodb://localhost/
2026-05-05T16:36:19.885+0000 1401 document(s) imported successfully. 0 document(s) failed to import.
```

e. The output of each query:

Query A: Indexing

```

moviesDB> db.moviesColl.createIndex({ name: 1 });
name_1
moviesDB> db.moviesColl.find({ name: "Ice Airport Alaska" });
[
  {
    _id: ObjectId('69fa168dd5e241a8acc65c9b'),
    link: 'https://www.amazon.com/gp/video/detail/amzn1.dv.gti.fcbadfca-e87f-9adb-a197-9f951e3117bb',
    name: 'Ice Airport Alaska',
    description: 'A chronicle of the daily challenges of the airport staff who manage to keep the Ted Stevens Anchorage International Airport (ANC) open, freight flowing, and passengers flying, all despite weather that regularly dips to -38 Fahrenheit and wind speeds that typically exceed 100 mph. Ice Airport Alaska has one or more episodes streaming on Paramount+, streaming with subscription on FuboTV, streaming with subscription on Smithsonian Earth (Via Prime Video), and 3 others. It's a documentary show with 12 episodes over 2 seasons. Ice Airport Alaska is still ailing with no announced date for the next episode or season. It has a high IMDb audience rating of 7.8 (48 votes).',
    released_at: '2020-10-26',
    genre: 'Documentary',
    poster: 'https://img.reelgood.com/content/show/0045f9a9-943b-4dda-9a01-384d5bc51f3d/poster-780.jpg',
    streaming_on: 'Paramount+, Prime Video',
    country: '',
    number_of_seasons: '2',
    type: 'TV Show',
    content_rating: '',
    imdb_rating: '7.8/10',
    cast_and_crew: '{}',
    uniq_id: '39e5ebf0-3ccd-52e0-a9ed-79dcd7ba0591',
    scraped_at: '02/07/2022 20:36:09'
  }
]

```

Query B: Logical Operations

```

moviesDB> db.moviesColl.find((type: 'Movie', $or: [{ genre: 'Drama' }, { genre: 'Documentary' } ]))
[
  {
    _id: ObjectId('69fa168dd5e241a8acc65c9c'),
    link: 'https://www.amazon.com/gp/video/detail/amzn1.dv.gti.28092bc-81e8-8a17-99e5-bdd6c9c00279',
    name: 'The Gift of Acadia',
    description: 'The Gift of Acadia is free on Tubi, and available for rent or purchase on Prime Video. It's a documentary movie.',
    released_at: '',
    genre: 'Documentary',
    poster: 'https://img.reelgood.com/content/movie/ada7bd12-f089-418a-b0fa-d0208d492a7a/poster-780.jpg',
    streaming_on: 'Free Services, Prime Video',
    country: '',
    number_of_seasons: '',
    type: 'Movie',
    content_rating: '',
    imdb_rating: '',
    cast_and_crew: '{}',
    uniq_id: '46bc21db-1200-413b-9f6c-6c466d27018',
    scraped_at: '02/07/2022 20:36:11'
  },
  {
    _id: ObjectId('69fa168dd5e241a8acc65c9d'),
    link: 'https://www.amazon.com/gp/video/detail/amzn1.dv.gti.29a81595-29d4-41f8-8dc3-d5ef708cda9',
    name: 'Mike Epps: Under Rated & Never Faded',
    description: 'Stand-up performance filmed at the historic Fox Theatre in Detroit, MI. Epps gets the house rocking with his unique and hilarious observations of married men, black/white family dynamics and a spot-on impersonation of a popular crime scene investigator. Comedy hip hop master Doug E. Fresh also makes a special appearance. Mike Epps: Under Rated & Never Faded featuring Mike Epps and Doug E. Fresh is streaming with subscription on Prime Video, and available for rent or purchase on VOD. It's a comedy and Documentary movie with an average IMDb audience rating of 6.2 (234 votes).',
    released_at: '2007-11-16',
    genre: 'Comedy, Documentary',
    poster: 'https://img.reelgood.com/content/movie/002a7c36-c5db-43db-9041-479ab82cbb0b/poster-780.jpg',
    streaming_on: 'Prime Video, Prime Video',
    country: '',
    number_of_seasons: '',
    type: 'Movie',
    content_rating: '',
    imdb_rating: '6.2/10',
    cast_and_crew: '({(name: 'Marcus Raboy', 'description': 'Director'), (name: 'Mike Epps', 'description': 'actor'), (name: 'Doug E. Fresh', 'description': 'actor'), (name: 'Mike Epps', 'description': 'author')})',
    uniq_id: '96620422-ebdb-5fca-8cde-1c35772a0480',
    scraped_at: '02/07/2022 20:36:09'
  },
  {
    _id: ObjectId('69fa168dd5e241a8acc65c9f'),
    link: 'https://www.amazon.com/gp/video/detail/amzn1.dv.gti.af4880ba-5bd-459c-8992-17a0af5da272',
    name: 'Santa Clara',
    description: 'Sammy, a man in his 40's works as a cowboy, lives in the outsidies of remote village in Beni. His past is unclear, rumors say that he's a fugitive, an assassin or a bandit. A work offer of driving a 2000 cattle herd to the Santa Clara ranch that is a really far comet now day. Is a well-paid job offer, and he's the only one that can do it, he needs the money, but going to Santa Clara will force him to leave his past behind or finally take revenge from those who forced him to the exile. Santa Clara is available for rent or purchase on Prime Video. It's a drama movie with an average IMDb audience rating of 6.2 (25 votes).',
    released_at: '2019-12-25',
    genre: 'Drama',
    poster: 'https://img.reelgood.com/content/movie/7802db-43be-4db7-0072-af22c95cc283/poster-780.jpg',
    streaming_on: 'Prime Video',
    country: '',
    number_of_seasons: '',
    type: 'Movie',
    content_rating: ''
  }
]

```

Query C: Aggregation

```

moviesDB> db.moviesColl.aggregate([ { $group: { _id: "$type", count: { $sum: 1 } } }, { $sort: { count: -1 } } ] );
[ { _id: 'Movie', count: 1232 }, { _id: 'TV Show', count: 169 } ]

```

Query D: Query with Sorting and Limiting

```

moviesDB> .editor
// Entering editor mode (Ctrl+D to finish, Ctrl+C to cancel)
db.moviesColl.find({
  type: "Movie",
  released_at: { $gte: "2011-01-01" },
  streaming_on: /Prime Video|Netflix/i,
  imdb_rating: { $ne: "" }
}), {
  _id: 0,
  name: 1,
  released_at: 1,
  genre: 1,
  streaming_on: 1,
  country: 1,
  imdb_rating: 1
}).sort({ imdb_rating: -1 }).limit(5);
[
  {
    name: "Attache",
    released_at: '2018-08-25',
    genre: 'Documentary',
    streaming_on: 'Prime Video ',
    country: '',
    imdb_rating: '9/10'
  },
  {
    name: 'Liquor House Comedy presents Brown Sugar Night',
    released_at: '2020-04-24',
    genre: 'Documentary',
    streaming_on: 'Free Services, Prime Video',
    country: '',
    imdb_rating: '9.7/10'
  },
  {
    name: 'Messiah',
    released_at: '2016-04-23',
    genre: 'Drama, Horror',
    streaming_on: 'Prime Video ',
    country: '',
    imdb_rating: '9.3/10'
  },
  {
    name: 'Why She Smiles',
    released_at: '2021-03-01',
    genre: 'Biography, Documentary',
    streaming_on: 'Free Services, Prime Video',
    country: '',
    imdb_rating: '9.1/10'
  },
  {
    name: 'Tanzania Transit',
    released_at: '2018-05-03',
    genre: 'Animation, Documentary',
    streaming_on: 'Prime Video ',
    country: 'Netherlands',
    imdb_rating: '8/10'
  }
]
moviesDB>

```

BONUS: Advanced MongoDB Feature (Distinct)

```
moviesDB> db.moviesColl.distinct("country");  
[  
  '',  
  'Australia',  
  'Bolivia',  
  'China',  
  'Denmark',  
  'France',  
  'Hong Kong',  
  'India',  
  'Ireland',  
  'Japan',  
  'Malaysia',  
  'Netherlands',  
  'Peru',  
  'Portugal',  
  'Slovakia',  
  'Spain',  
  'Taiwan',  
  'Ukraine',  
  'Afghanistan',  
  'Austria',  
  'Brazil',  
  'Colombia',  
  'Egypt',  
  'Germany',  
  'Hungary',  
  'Indonesia',  
  'Israel',  
  'Kazakhstan',  
  'Mexico',  
  'New Zealand',  
  'Philippines',  
  'Romania',  
  'South Africa',  
  'Sweden',  
  'Thailand',  
  'United Kingdom',  
  'America',  
  'Belgium',  
  'Canada',  
  'Czechia',  
  'Ethiopia',  
  'Greece',  
  'Iceland',  
  'Iran',  
  'Italy',  
  'Kiribati',  
  'Morocco',  
  'Norway',  
  'Poland',  
  'Russia',  
  'South Korea',  
  'Switzerland',  
  'Turkey',  
  'Venezuela'  
]
```